

AI Restaurant Voice Assistant

End-to-end n8n backend for an ElevenLabs phone-call voice agent — technical & process documentation

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This report documents the n8n backend that powers an ElevenLabs Conversational AI voice agent for a restaurant. Callers can check table availability, book a reservation, place a pickup or delivery order, cancel or modify an existing reservation, or ask for a human callback — entirely by voice — while n8n validates every request, keeps a Google Sheets record, and keeps the manager notified by email.

1. Executive Summary

Restaurant phone lines get flooded with the same handful of requests: “do you have a table for 4 at 8?”, “can I get a pickup order in?”, “I need to cancel my booking”. This backend gives an ElevenLabs voice agent five callable tools that handle exactly these requests, each backed by real capacity checking and validation so the agent never overbooks the dining room, orders never go unrecorded, and reservation changes always match an existing booking before anything is altered.

Metric	Detail
Trigger	5 webhooks called directly by ElevenLabs during a live phone call
Voice platform	ElevenLabs Conversational AI — the caller talks to an agent, which calls these tools mid-conversation
Data store	Google Sheets — Reservations, orders and Escalations tabs
Manager notifications	Gmail — sent on every new booking, order, cancellation, modification and escalation
Capacity model	Fixed table capacity (40 covers) checked per date/time slot before a booking is confirmed
Reservation lookup	Cancel/modify requests are matched by phone number + date before any change is made

2. Business Value

- Phone lines never busy — reservations, orders and availability checks are handled by voice at any hour, without tying up staff.
- No overbooked dining room — every reservation is checked against real capacity for that exact date and time before it's confirmed.
- Orders never lost in translation — pickup and delivery orders are logged verbatim to the sheet and emailed to the kitchen/manager immediately.
- Safe changes only — a reservation can only be cancelled or modified once it's matched by phone number and date, so the agent can't alter the wrong booking.
- Manager always informed — every booking, order, cancellation, modification and callback request lands in their inbox with full detail in real time.

3. Architecture Overview

The ElevenLabs voice agent is configured with five Webhook Tools, each pointing at its own n8n webhook. During a call, the agent's LLM decides which tool to call based on what the caller is asking for, extracts the parameters from the conversation, and reads the JSON response back to the caller.

Tool / Webhook	Purpose
check_availability	Sums booked party sizes for the requested date/time and compares against total table capacity
make_reservation	Saves a confirmed booking to Reservations and emails the manager
place_order	Saves a pickup or delivery order to the orders tab and emails the manager
manage_reservation	Finds a reservation by phone + date, then cancels (deletes the row) or modifies it
request_human_callback	Logs an escalation and sends an urgent alert to the manager

4. Tool — check_availability

- 1 Capture the request**
The requested date, time and party size are taken from the caller's request.
- 2 Read existing reservations**
Every row is pulled from the Reservations sheet so today's actual bookings can be summed up.
- 3 Check real capacity**
Dates and times are normalised into a consistent format, then every non-cancelled reservation for that exact date and time is totalled up by party size and compared against a fixed table capacity of 40 covers; the requested party is accepted only if it fits in what's left.
- 4 Reply in natural language**
The agent gets a clear yes/no plus a spoken-friendly message — confirming the slot works, or apologising that it's full and asking if an earlier or later time would work instead.

5. Tool — make_reservation

- 1 Capture the booking**
Customer name, phone number, party size, date, time and any special requests (defaulting to “None”) are collected from the caller.
- 2 Save & notify**
The booking is appended to the Reservations sheet, and a formatted email goes to the manager with the guest, phone, party size, date, time and any special requests.
- 3 Confirm back to the caller**
The agent reads back a confirmation naming the guest, party size, date and time so the caller leaves the call certain their table is booked.

6. Tool — place_order

1 Capture the order

Customer name, phone number, the items ordered, order type (pickup or delivery), and a delivery address when relevant (defaulting to “N/A” for pickup) are collected.

2 Save & notify

The order is appended to the orders sheet, and the manager/kitchen receives a formatted email with the full item list, order type and delivery address.

3 Confirm back to the caller

The caller is told their order was received and that a team member will confirm the total and estimated time shortly — pricing and timing are handled by a human, not guessed by the AI.

7. Tool — manage_reservation

1 Find the matching reservation

The Reservations sheet is searched for a non-cancelled booking whose phone number and date both match what the caller provided — this double check prevents altering the wrong guest's reservation.

2 Not found? Stop safely

If nothing matches both the phone number and the date, the caller is asked to double-check their details instead of the agent guessing which booking they meant.

3 Cancel

For a cancel action, the matching row is deleted from the sheet entirely, and the manager is emailed a cancellation notice with the original booking details struck through for clarity.

4 Modify

For a modify action, only the fields the caller actually changed (new date, time or party size) are merged over the existing booking, the row is updated in place, and the manager is emailed the new details.

5 Confirm back to the caller

The agent receives a status of cancelled, modified or not_found along with a message it can speak directly to the caller.

8. Tool — request_human_callback

1

Capture the request

Customer name, phone number and the reason they need a human are recorded, along with the conversation ID for traceability.

2

Log & alert the manager

The request is appended to the Escalations tab, and the manager receives an urgent-styled email with the customer's details, the reason, and a one-tap call link.

3

Confirm back to the caller

The caller is told the request has been flagged for the team and someone will call them back shortly — this tool never transfers the live call, it only queues a follow-up.

9. Node-by-Node Map

Tool	Key nodes
check_availability	Availability Webhook → Read Reservations → Check Capacity → Is Available → Message - Available / Not Available → Respond Availability
make_reservation	Reservation Webhook → Normalize Reservation Data → Log Booking to Sheet → Build Confirmation Message / Send Manager New Reservation Email → Respond to ElevenLabs
place_order	Order Webhook → Normalize Order Data → Log Order to Sheet → Build Order Confirmation / Send Manager New Order Email → Respond Order
manage_reservation	Manage Reservation Webhook → Read Reservations (for lookup) → Find Matching Reservation → Reservation Found → Is Cancel Action → Delete Reservation Row / Merge Modify Fields with Fallback → Update Reservation Details → Respond Manage
request_human_callback	Escalation Webhook → Normalize Escalation Data → Log Escalation to Sheet → Send Manager Escalation Email / Build Escalation Response → Respond Escalation

10. Reliability & Security Design

- **Real capacity checking** — availability is calculated from actual booked party sizes for that exact slot against a fixed table capacity, not left to the AI to estimate.
- **Phone + date matching for changes** — cancelling or modifying a reservation requires both the phone number and the date to match an existing booking, preventing accidental changes to the wrong guest's table.
- **Graceful “not found” handling** — when a cancel/modify lookup fails, the caller is asked to double-check details rather than the system guessing or failing silently.
- **Pricing kept out of the AI's hands** — orders are logged as received, with total and timing confirmed by a human, so the agent never quotes a price or ETA it can't guarantee.
- **Full audit trail** — every reservation, order, cancellation, modification and escalation is logged to Google Sheets and emailed to the manager, so nothing happens invisibly.

11. Setup Checklist

- Connect a Google Sheets credential and point every Sheets node at your own spreadsheet, with Reservations, orders and Escalations tabs.
- Connect a Gmail credential and replace the manager@example.com placeholder on the five “Send Manager ...” email nodes with your real notification address.
- Set the restaurant's real table capacity in the Check Capacity code node (currently 40 covers) to match your venue.
- Activate the workflow, then copy each webhook's Production URL into the matching ElevenLabs Webhook Tool: check_availability, make_reservation, place_order, manage_reservation, request_human_callback.
- Test all five tools end-to-end with a live or simulated call: check availability, book, place an order, modify, cancel, and request a callback.

This workflow was designed and built end-to-end in n8n as part of an AI automation portfolio, demonstrating a voice-agent backend for a restaurant with real-time capacity checking, order logging and safe reservation management.